

Select a learning theory that you feel most closely relates to your approach to teaching. Compare and contrast this theory to others that have been discussed. How does that relate to teaching online courses? What aspects of the theory do you feel best apply to learning?

After reflecting on various learning theories including those of behaviorism, cognitivism, and constructivism, I feel that cognitivism best reflects my personal approach to teaching. More specifically, social cognitive theory (first described by Bandura in the 1960s) best defines my educational approach with medical students and trainees.

Social cognitive theory incorporates the triad of personal factors, behavior, and environmental factors to influence learning through a construct coined Reciprocal Determinism or Triadic Reciprocity Causation as described by Albert Bandura. In medical education, I feel this best reflects my learning style through its adaptability, and ability to leverage prior clinical cases or experiences to target learning concepts. For example, let's say I wish to teach a group of medical trainees about hepatitis A. Simply reading from a textbook and attempting rote memorization is one mechanism for a student to learn about this virus and its clinical manifestation. I find observational learning to be a particularly effective strategy for medical trainees, augmenting book learning with live clinical experiences. This also allows for opportunities for modeling, demonstrating to trainees how to incorporate clinical guidelines (or best practice recommendations) in real time. I can draw from my prior experiences managing patients with hepatitis A who may have suffered from atypical or rare complications and use these experiences to augment my teaching. In particular, I appreciate the adaptability of modeling in clinical education, leveraging tools such as attention (during clinical rounds) and motivation (discussing cost-effective medical care, or potential side effects of therapy that could lead to patient harm). These tools not only serve as educational for trainees, but also for patients.

Other facets of Bandura's reciprocal determinism are described below, when compared against other learning theories.

Comparing Bandura's theories against behavioral learning theory, I find behaviorism less suited towards my approach in medical education. One such example is that of operant conditioning as an educational tool given the potential for patient harm in the process. For example, when teaching trainees about endoscopy, one important technique that I teach is loop reduction. When performing colonoscopies, there are often twists and turns that must be navigated using the scope. Given the camera is affixed at the end of the device, you cannot visualize the scope that is behind the forward-facing camera but is still inside the patient. When navigating the colon, one sometimes encounters areas where one is inserting more scope into a patient; however, the tip of the endoscope does not advance further and in fact can stay stationary or travel retrograde. The term for this is 'looping' and if a trainee does not recognize this, not only can it cause a patient to experience pain or discomfort, the pressure exerted by a colonoscope loop can lead to splenic injury (such as a laceration) or colonic perforation. Both of these significant complications can lead to hemodynamic instability, need for emergent surgery, or even mortality. As such, negative reinforcement would not be an effective educational strategy in endoscopy given the risk for patient harm. Though positive reinforcement can be utilized to celebrate a learner's achievements (such as reaching the cecum, or successful polypectomy), modeling I find is a much more effective educational tool. Through the three goal properties described by Schunk (2020), namely specificity, proximity, and difficulty, endoscopy can be a much more achievable skill to finesse during training. I personally utilize specificity to set the expectation with trainees prior to starting an endoscopy so that if they have difficulties during the procedure, they are less discouraged overall. As well, they have the ability to learn via modeling

if I need to take over during a procedure to aid during a difficult segment. A colonoscopy can be split into several specific goals, such as advancing the scope past a given point (the sigmoid colon is often challenging to navigate for trainees), intubating the cecum (the goal anatomic landmark in colonoscopy), polyp identification, and minimizing looping. By highlighting specific goals within a procedure for a trainee to achieve, this often can be a more effective educational strategy. Proximity goals can include time efficiency during endoscopy (not running late in a morning block of cases). These goals would be much more challenging to achieve using behaviorism, such as emotional conditioning or the use of desensitization (as described by Pavlov).

Vygotsky's theories on social constructivism, especially incorporating the zone of proximal development can be utilized in medical education; however, I find it less effective overall in my own practice. Though the reciprocal teaching (Schunk 2020) can be leveraged as a means of incorporating social interaction and the use of scaffolding can augment comprehension on a given topic, I find these strategies more challenging to incorporate in a clinical setting as I feel this would be a rather time intensive process on the whole. Though incorporating social interaction can be used as a mentoring tool (such as through cognitive apprenticeships), this is perhaps a better representation of fellowship as a training mechanism in a big picture perspective. Though in clinical interactions with patients or more focused medical education I find this less useful as an educational tool.

I often have incorporated elements of information processing theory in medical education, as a means to augment teaching that can happen at a patient's bedside. The dual memory model incorporates short- and long-term memory retention, and by leveraging coding as a type of control process I try to add pertinent information when seeing a patient with a given

medical condition. Using the example of a patient with a hepatitis A infection as discussed earlier, I often use an opportunity such as this at a patient's bedside to not only provide more information for the patient regarding their affliction, this is also an educational opportunity to teach the importance of vaccinating close contacts (both important for the patient, and a learning lesson for medical trainees) as well as potential future complications of their infection. The added stimulus of providing a live clinical example for trainees to recall when trying to learn about hepatitis A, can be a powerful educational tool. Social cognitive theory incorporates similar elements towards my educational strategy, as I am leveraging the environment (namely, the patient with the affliction) as a tool to augment trainee education and motivate them to learn more about the underlying condition following medical rounds. My hope through this strategy is for the trainee to gain self-efficacy so that they can reinforce the concepts learned when following up with the patient the following day (or during a future clinic visit) or when dealing with patients they encounter in the future also with hepatitis A infection.

In relation to teaching online courses, I believe Bandura's theory to be a particularly effective educational tool. By understanding the virtual environment in online learning, this can be an important tool in motivating learners and in modeling behavior. For example, say a teacher is utilizing Zoom or Microsoft Teams to provide a lecture utilizing PowerPoint. Educators should model effective use of slides, avoid extraneous information else it may overload auditory and visual domains (an example of cognitive overload as discussed by Mayer & Moreno in 2003). Students can be assigned concepts to self-study and observe examples of effective PowerPoint presentations, and they can then imitate those same effective educational strategies when presenting remotely to the rest of the class. The ability of asynchronous learning through online media is another benefit to online courses. This allows learners to better control their

environment for one that is more conducive to their learning style, and motivated learners can also incorporate self-regulation. Self-regulation incorporates three components, namely self-observation judgement, and self-response (Schunk 2020). Artificial intelligence (AI) in the future can be a tool to be incorporated by educators when teaching online classes, such as incorporating large-language model derived chatbots (such as ChatGPT or Google Bard) could augment existing tools such as discussion boards. Though it would still be important to have human responses to a student's post, incorporating an AI-augmented chatbot could be an effective means of replying to students whose posts had fewer human replies (potentially due to submission close to a deadline). Students could then be afforded equal opportunities to reply to these comments (otherwise without AI assistance, there may be a discrepancy in the number of replies per comment that would need to be addressed). These chatbots could also be used to aid with teacher grading (comparing a student's submission against a provided rubric). Going back to Bandura's model, knowing that there may be AI-aided grading or an AI component to a course could lead to increased attention spent on a submission rubric (to ensure all components are incorporated or addressed), and serve as added motivation to ensure submissions are clear and well-written (else AI may have difficulties interpreting their responses). Though the technology is very early at this time and not in a position for immediate integration, it is important for educators to keep up with technological advances as I suspect AI will become an ever-growing component of future online learning and teaching.

Mentoring is an important concept that leverages multiple facets of social cognitive theory (Schunk 2020) including modeling (an important part of Bandura's reciprocal determinism), and online tools can be leveraged to augment the relationship between a mentor and a mentee and should be incorporated in online teaching. No longer do in-person educational

sessions need to take place, but mentors can leverage their connections across the country (or across the world) to link mentees with colleagues that would have otherwise been inaccessible leading to unique educational opportunities. As technology continues to advance, online live translation tools continue to improve and will soon allow for reliable means of connecting teachers (or mentors) with students across the globe and remove language as a potential barrier towards learning. Having access to these new educational resources could serve as a motivational strategy for learners and augment observational learning by removing language as a barrier (modulating the learning environment) and allowing for cross-cultural experiences and role models. Online mentoring sessions could also be recorded to allow for a mentee to reflect on prior conversations and improve retention. These new learning modalities could instill unique critical thinking and problem-solving skills (Schunk & Mullen, 2013) that otherwise would not be feasible outside of online teaching opportunities. These are all important facets for educators to consider when creating online curricula or designing online courses, and to incorporate into their teaching strategies.

References:

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